

Air University



Programming Fundamentals (Lab)

Year 2024

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Academic Year: 2024 – 2028

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BS – Cyber Security – Fall – 24 (A)

Date of Submission: November 13, 2024

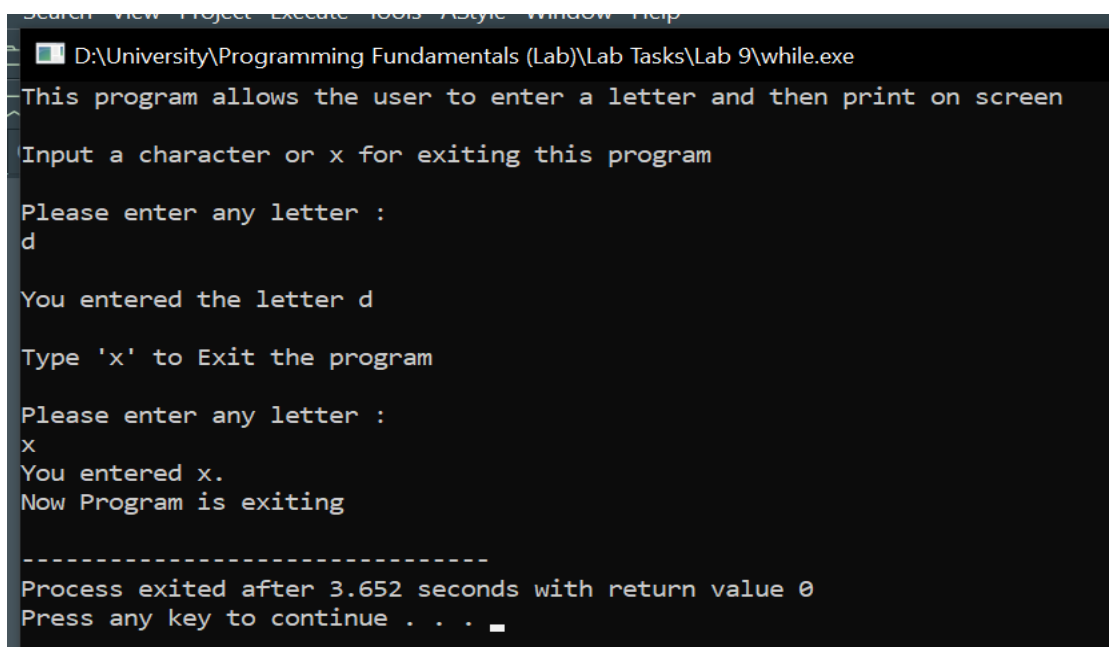
Lab # 09

Lab 9.1A – while loop

Code

```
1 #include <iostream>
2
3 using namespace std;
4
5 int main()
6 {
7     char letter = 'a';
8
9     cout<<"This program allows the user to enter a letter and then print on screen"<<endl;
10    cout<<"\nInput a character or x for exiting this program"<<endl;
11
12    while (true)
13    {
14
15        cout<<"\nPlease enter any letter : " << endl;
16        cin>>letter;
17
18        if (letter == 'x')
19        {
20            cout << "You entered x. \nNow Program is exiting" << endl;
21            break;
22        }
23
24        cout << "\nYou entered the letter " << letter << endl;
25        cout << "\nType 'x' to Exit the program " << endl;
26    }
27    return 0;
28 }
```

Output



```
Search View Project Execute Tools Style Window Help
D:\University\Programming Fundamentals (Lab)\Lab Tasks\Lab 9\while.exe
This program allows the user to enter a letter and then print on screen
Input a character or x for exiting this program
Please enter any letter :
d
You entered the letter d
Type 'x' to Exit the program
Please enter any letter :
x
You entered x.
Now Program is exiting
-----
Process exited after 3.652 seconds with return value 0
Press any key to continue . . .
```

Lab 9A – Task 1

Code

```
1 #include <iostream>
2 using namespace std;
3 int main()
4 {
5     //variable declaration
6     int num, numFact=1 ;
7
8     //taking input from user
9     cout << "Enter number for Factorial " << endl;
10    cin >> num ;
11
12    //checking validation and calculating factorial
13    if (num > 0)
14    {
15        while (num > 0)
16        {
17            numFact = numFact * num ;
18            num -- ;
19        }
20        cout << "The Factorial of number is " << numFact << endl;
21    }
22    else
23    {
24        cout << "Kindly Enter Positive number " << endl;
25    }
26    return 0;
27 }
```

Output: 1

```
(globals)
D:\University\Programming Fundamentals (Lab)\Lab Tasks\Lab 9\Lab 9A - Task 1.exe
Enter number for Factorial
-6
Kindly Enter Positive number

-----
Process exited after 3.506 seconds with return value 0
Press any key to continue . . .
```

Output: 2

```
D:\University\Programming Fundamentals (Lab)\Lab Tasks\Lab 9\Lab 9A - Task 1.exe
Enter number for Factorial
5
The Factorial of number is 120

-----
Process exited after 1.65 seconds with return value 0
Press any key to continue . . .
```

Lab 9A – Task 2

Code

```
1  #include <iostream>
2
3  using namespace std;
4  const int NUMBEROFTTESTS = 5;
5
6  int main()
7  {
8
9      int score, test, num=1 ; // the individual score read in
10     float total = 0.0; // the total of the scores
11     float average; // the average of the scores
12
13     while ( num <=5 )
14     {
15         cout << "\nEnter your score on test " << num << ": " << endl;
16         cin >> score ;
17
18         if ( score >= 0 && score <= 100)
19         {
20
21             total += score ;
22
23             num ++;
24         }
25         else
26         {
27             cout << "\nKindly Enter Value ranging from (0 - 100) " << endl;
28         }
29     }
30
31     //calculate the average of 5 scores
32     average = total / 5 ;
33     cout << "Your average based on " << NUMBEROFTTESTS
34     << " test scores is " << average << endl;
35     return 0;
36 }
```

Output: 1

```
D:\University\Programming Fundamentals (Lab)\Lab Tasks\Lab 9\MarksEntry.exe
Enter your score on test 1:
98
Enter your score on test 2:
78
Enter your score on test 3:
0
Enter your score on test 4:
-1
Kindly Enter Value ranging from (0 - 100)
Enter your score on test 4:
56
Enter your score on test 5:
67
Your average based on 5 test scores is 59.8
-----
Process exited after 18.57 seconds with return value 0
Press any key to continue . . .
```

Output: 2

```
D:\University\Programming Fundamentals (Lab)\Lab Tasks\Lab 9\MarksEntry.exe
Enter your score on test 1:
56
Enter your score on test 2:
67
Enter your score on test 3:
78
Enter your score on test 4:
89
Enter your score on test 5:
90
Your average based on 5 test scores is 76
-----
Process exited after 7.005 seconds with return value 0
Press any key to continue . . .
```

Lab 9.1B – Running Total

Code

```
1 // This program takes daily sales figures over a period of time
2 // and calculates their total.
3 #include <iostream>
4 #include <iomanip>
5 using namespace std;
6
7 int main()
8 {
9     double total = 0.0; // Accumulator, initialized with 0
10    int days;
11    float sale;
12    // Get the number of days.
13    cout << "Enter the number of days " << endl;
14    cin >> days ;
15
16    // Get the sales for each day and accumulate a total.
17
18    int i=1;
19
20    while( i <= days) //complete the expression
21    {
22        //input sale figure of the day
23        cout << "Enter the sales for day " << i << endl; //....
24        cin >> sale;
25
26        // Accumulate the running total.
27        total += sale;
28        //update counter variable
29        i++;
30    }
31
32    // Display the total sales.
33    cout << fixed << showpoint << setprecision(2)<<endl;
34    cout << "The total sales are $" << total << endl;
35    return 0;
36 }
```

Output:

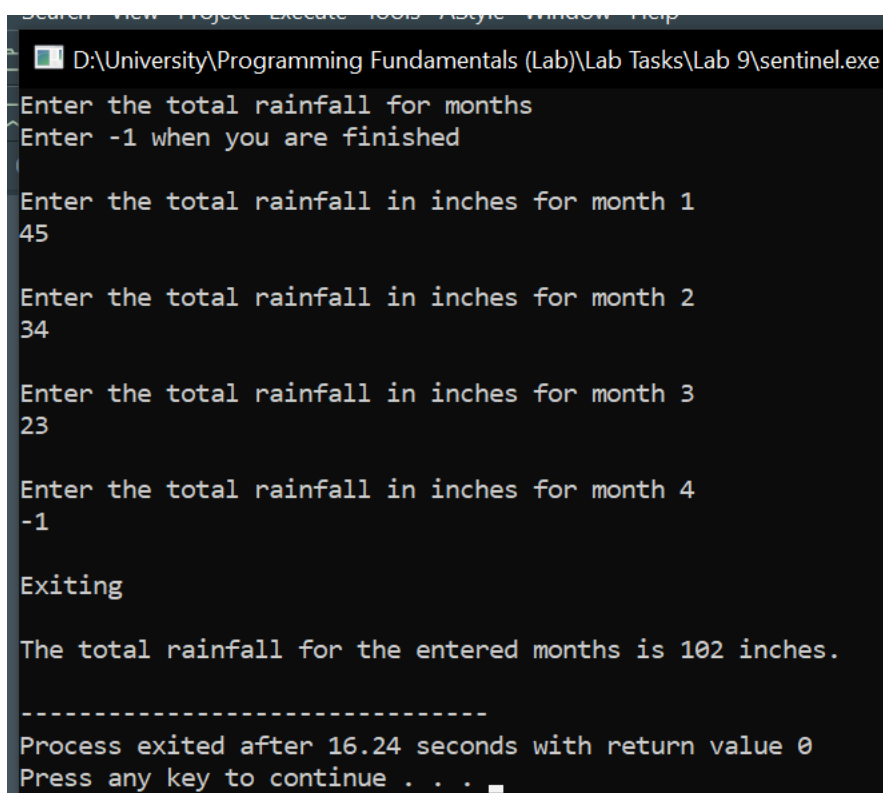
```
D:\University\Programming Fundamentals (Lab)\Lab Tasks\Lab 9\RunningTotal.exe
Enter the number of days
7
Enter the sales for day 1
98.75
Enter the sales for day 2
88.70
Enter the sales for day 3
80.35
Enter the sales for day 4
75.40
Enter the sales for day 5
80.25
Enter the sales for day 6
40.70
Enter the sales for day 7
95.70

The total sales are $559.85

-----
Process exited after 52.83 seconds with return value 0
Press any key to continue . . .
```

Lab 9.2B – Sentinel**Code**

```
1 // This program illustrates the use of a sentinel in a while loop.
2 // The user is asked for monthly rainfall totals until a sentinel
3 // value of -1 is entered. Then the total rainfall is displayed.
4 // USAMA ARSHAD
5
6 #include <iostream>
7 using namespace std;
8
9 int main()
10 {
11     //define and initialize to 1 the variable month
12     float total = 0, rain;
13     int month = 1;
14
15     cout << "Enter the total rainfall for months " << endl;
16     cout << "Enter -1 when you are finished" << endl;
17
18     // while rain does not equal -1
19     while (rain != -1)
20     {
21         cout << "\nEnter the total rainfall in inches for month "
22         << month << endl;
23         cin >> rain;
24
25         if (rain > 0 )
26
27             total += rain ;
28
29     }
30
31     else if ( rain = -1 )
32
33         cout << endl << "Exiting " << endl ;
34
35     else
36
37         cout << "Kindly Enter Valid rainfall in positive number" << endl;
38
39         month ++ ;
40
41     }
42
43     if ( month == 1 )
44
45         cout << "No data has been entered" << endl;
46
47     else
48
49         cout << "\nThe total rainfall for the entered months is "<< total << " inches." << endl;
50
51     return 0;
```

Output:

```
Search View Project Execute Tools Format Window Help
D:\University\Programming Fundamentals (Lab)\Lab Tasks\Lab 9\sentinel.exe
Enter the total rainfall for months
Enter -1 when you are finished

Enter the total rainfall in inches for month 1
45

Enter the total rainfall in inches for month 2
34

Enter the total rainfall in inches for month 3
23

Enter the total rainfall in inches for month 4
-1

Exiting

The total rainfall for the entered months is 102 inches.

-----
Process exited after 16.24 seconds with return value 0
Press any key to continue . . .
```


ATMCode

```
1  #include <iostream>
2  using namespace std;
3  int main()
4  {
5      //declaring variables
6      int act, choice , balance = 10000, amount;
7      do
8      {
9          cout << "\nChoose Action" << endl;
10         cout << "\n1. Balance \n2. Withdraw \n3. Deposit \n4. Exit " << endl;
11         cout << "\nEnter Your Action " << endl;
12         cin >> act;
13
14         switch(act)
15         {
16             case 1 :
17                 cout << "Your Balance is " << balance << endl;
18                 break;
19
20             case 2 :
21                 cout << "Enter amount for Withdrawal " << endl;
22                 cin >> amount;
23
24                 if (amount > balance)
25                     cout << "Insufficient Balance" << endl;
26
27                 else if (amount % 500 == 0)
28                 {
29
30                     balance = balance - amount;
31                     cout << "Remaining Balance is " << balance << endl ;
32                 }
33                 else
34                 {
35                     cout << "kindly Enter multiple of 500 " << endl;
36                 }
37                 break;
38
39             case 3 :
40                 cout << "Enter Amount for Deposit " << endl;
41                 cin >> amount;
42                 balance = balance + amount;
43                 cout << "New Balance is " << balance << endl;
44                 break;
45
46             case 4 :
47                 cout << "Signing Off" << endl;
48                 break;
49             default:
50                 cout << "Kindly Enter Valid choice " << endl;
51         }
52     }
53     while ( act != 4 ) ;
54     return 0;
55 }
```

Output

```
D:\University\Programming Fundamentals (Lab)\Activity\ATM.exe
Choose Action
1. Balance
2. Withdraw
3. Deposit
4. Exit
Enter Your Action
1
Your Balance is 10000
Choose Action
1. Balance
2. Withdraw
3. Deposit
4. Exit
Enter Your Action
2
Enter amount for Withdrawal
1300
Kindly Enter multiple of 500
Choose Action
1. Balance
2. Withdraw
3. Deposit
4. Exit
Enter Your Action
2
Enter amount for Withdrawal
5000
Remaining Balance is 5000
Choose Action
1. Balance
2. Withdraw
3. Deposit
4. Exit
Enter Your Action
3
Enter Amount for Deposit
5000
New Balance is 10000
Choose Action
1. Balance
2. Withdraw
3. Deposit
4. Exit
Enter Your Action
4
Signing Off
-----
Process exited after 39.03 seconds with return value 0
Press any key to continue . . .
```

Thank you for your Time!

The End